Comparison of Adaptive Gain Laws for newly Developed Rotor-Effective Wind Speed Estimator

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Why Wind Speed Estimation in FOWTs?

Why estimators matter?

- Region II: optimal torque & TSR tracking.
- Region III: pitch control & load mitigation.
- Feedforward & advanced control.

Key challenges:

- However, in real-world, direct and perfect measurement of Rotor-Effective Wind Speed (REWS) is impossible.
- Platform motions → strong nonlinearities.
- Turbulence + waves → modelling uncertainty.

Existing Solutions

Main approaches:

- LiDAR-based sensing.
- Learning-based methods.
- Observer-based methods:
  1- Extended Kalman Filter (EKF)
  2- Sliding-mode observers (SMO / SOSMO) ✓

1. Robustness
2. Stability Proof
3. Finite time convergence
4. Easy to implement

Limitations of LiDAR:
- Cost.
- Maintenance.
- Vulnerability of LiDAR measurements to motion-induced errors.

Limitations of Learning-based methods:
- Require offline training phase using large datasets.
- difficult to prove stability or robustness of CL systems.
- Unseen conditions.

Contributions

The observer-based model

$$\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u}) + \boldsymbol{\Delta}(t), \qquad y = h(\mathbf{x})$$

$$f(\mathbf{x}, \mathbf{u}) = \begin{bmatrix} \frac{1}{J}\left(\frac{\rho \pi R^3 v^2}{2\lambda} C_p(\lambda, \beta) - n_g \tau_g\right) & 0 \end{bmatrix}^{\top}$$

$$\boxed{\boldsymbol{\Delta}(t) = \begin{bmatrix} \delta(t) & f_v(t) \end{bmatrix}^{\top}}$$

$$\dot{\hat{\mathbf{x}}} = f(\hat{\mathbf{x}}, \mathbf{u}) + \left[\frac{\partial \boldsymbol{\Phi}}{\partial \hat{\mathbf{x}}}\right]^{-1} \begin{bmatrix} \boxed{-K_1} |\hat{\omega}_r - \omega_r|^{1/2} \operatorname{sign}(\hat{\omega}_r - \omega_r) \\ \boxed{-K_2} \operatorname{sign}(\hat{\omega}_r - \omega_r) \end{bmatrix}$$

Solutions (Adaptive Gains)

- Dynamically adjusts gains
- Prevents overestimation and mitigates chattering

Project Objective

Compare adaptive gain laws on the same observer, under the same wind/wave conditions

Key constraints

- Same Controller.
- Same measurements (Rotor speed ω).
- Same OpenFAST model (NREL 5 MW FOWT).

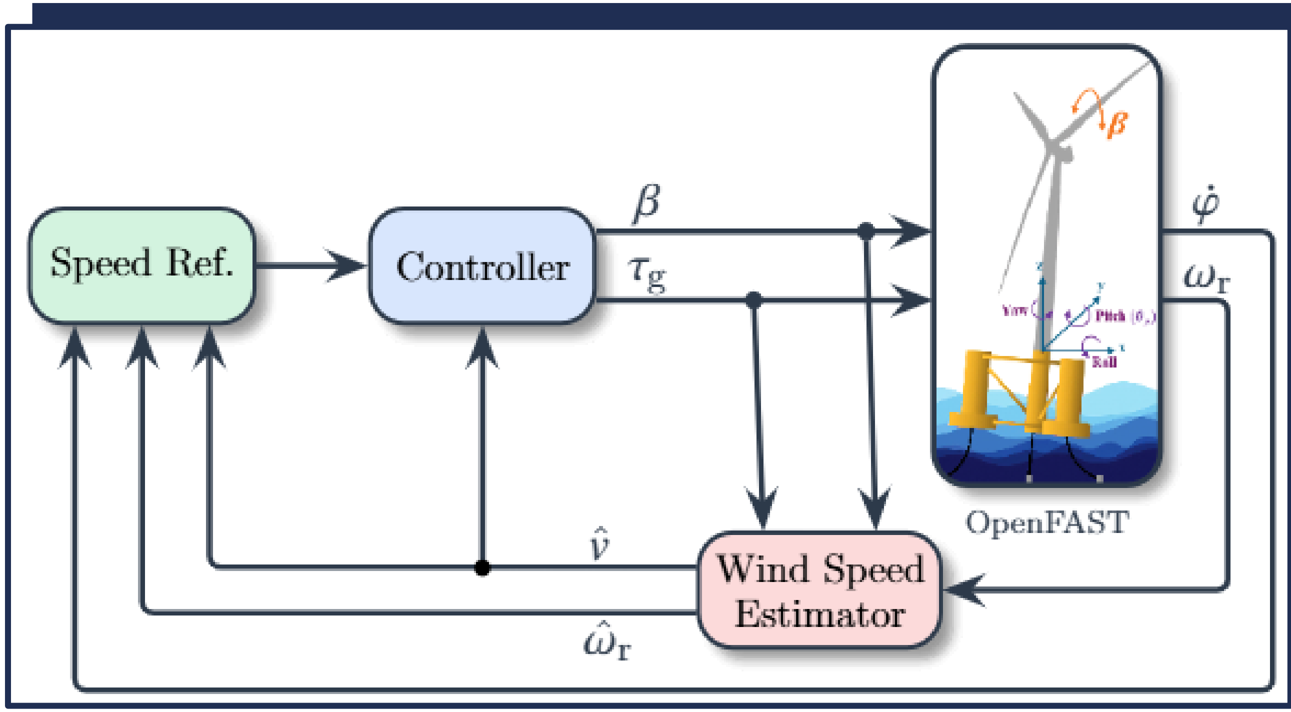
Comparing criteria

Accuracy

- Root Mean Square Error RMSE
- Variance of Estimation Error σ_e^2
- Mean Square Error E[e^2]

Complexity

- Number of tuning parameters



Model

Adaptive laws to compare

Shtessel et al. (2012) –6 parameters ASTW

6 parameter

Control Law:

$$\omega = -\alpha|\sigma|^{1/2}\text{sign}(\sigma) + v$$
$$\dot{v} = -\frac{\beta}{2}\text{sign}(\sigma)$$

Adaptive gains:

$$\begin{cases} \omega_1\sqrt{\frac{\gamma_1}{2}}\text{sign}(|\sigma| - \mu), & \text{if } \alpha > \alpha_m \\ \eta, & \text{if } \alpha \le \alpha_m \end{cases}$$

where $\varepsilon, \lambda, \gamma_1, \omega_1, \eta$ are arbitrary positive constants, and $\eta_1 \geq \mu$, $\eta_2 > 0$. The parameter α_m is an arbitrary small positive constant. □

Mirzaei et al. (2022) – Self-Tuning ASTW

1 parameter + Differentiator $\frac{s}{1+\tau s}$

Control Law:

$$u = -k_1|\sigma|^{1/2}\,\text{sgn}(\sigma) + v$$
$$\dot{v} = -k_2\,\text{sgn}(\sigma)$$

Adaptive gains:

$$\dot{k}_1 = \begin{cases} \frac{\alpha}{|\psi| + \varepsilon} & |\sigma| > \varepsilon \\ -k_1 & |\sigma| \le \varepsilon \end{cases}$$

$$\dot{k}_2 = \begin{cases} \frac{\alpha}{2|\sigma|^{1/2}} & |\sigma| > \varepsilon \\ -k_2 & |\sigma| \le \varepsilon \end{cases}$$

$$\psi = -\dot{\sigma}$$

Adaptive gains:

$$\dot{k}_1 = \begin{cases} \frac{\bar{\mu}}{|\psi| + \varepsilon} & |\sigma| > \varepsilon \\ -k_1 & |\sigma| \le \varepsilon \end{cases}$$

$$\dot{k}_2 = \begin{cases} \frac{\bar{\mu}}{2|\sigma|^{1/2}} & |\sigma| > \varepsilon \\ -k_2 & |\sigma| \le \varepsilon \end{cases}$$

$$\bar{\mu} = |k_1||\sigma|^{1/2} + |\psi| + |w|$$

$$\dot{w} = k_2\,\text{sgn}(\sigma)$$

Mirzaei et al. (2024) – Auto-Tuning ASTW

0 parameter + Differentiator $\frac{s}{1+\tau s}$

Control Law:

$$\bar{u}(t) = -k_1(t)|\sigma(t)|^{\frac{1}{2}}\,\text{sgn}\left(\sigma(t)\right) + w(t)$$
$$\dot{w}(t) = -k_2(t)\,\text{sgn}\left(\sigma(t)\right)$$

Adaptive gains:

$$\dot{k}_1(t) = \begin{cases} \frac{\alpha(t)}{|\psi|+\epsilon(t)} & \text{if } |\sigma(t)| > \epsilon(t) \\ -k_1(t) & \text{if } |\sigma(t)| \le \epsilon(t) \end{cases}, \quad k_1(0) = k_{01} > 0$$

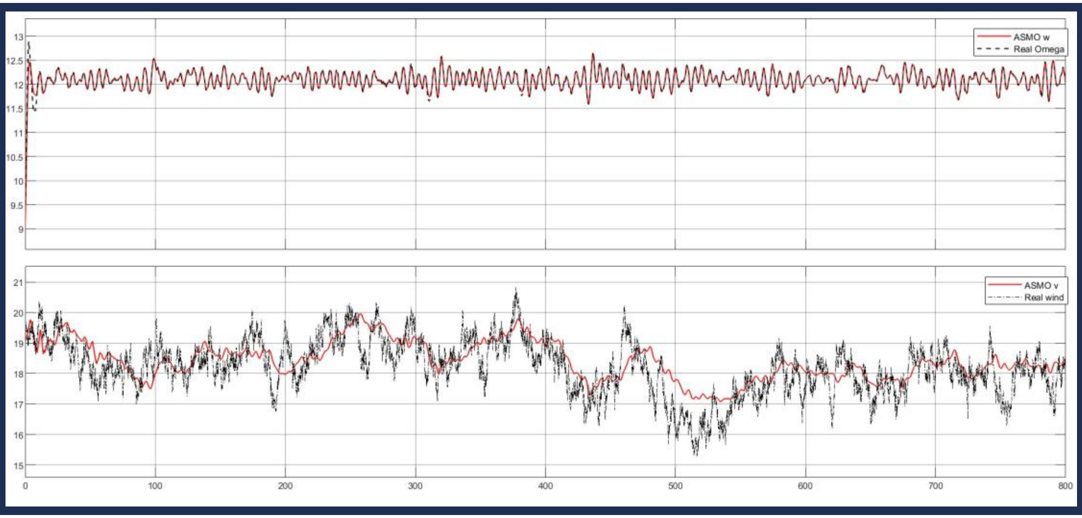
$$\dot{k}_2(t) = \begin{cases} \frac{\alpha(t)}{2|\sigma(t)|^{\frac{1}{2}}} & \text{if } |\sigma(t)| > \epsilon(t) \\ -k_2(t) & \text{if } |\sigma(t)| \le \epsilon(t) \end{cases}, \quad k_2(0) = k_{02} > 0$$

with $\psi = -\hat{\dot{\sigma}}(t)$, $\alpha(t)$ and $\epsilon(t)$ being defined as

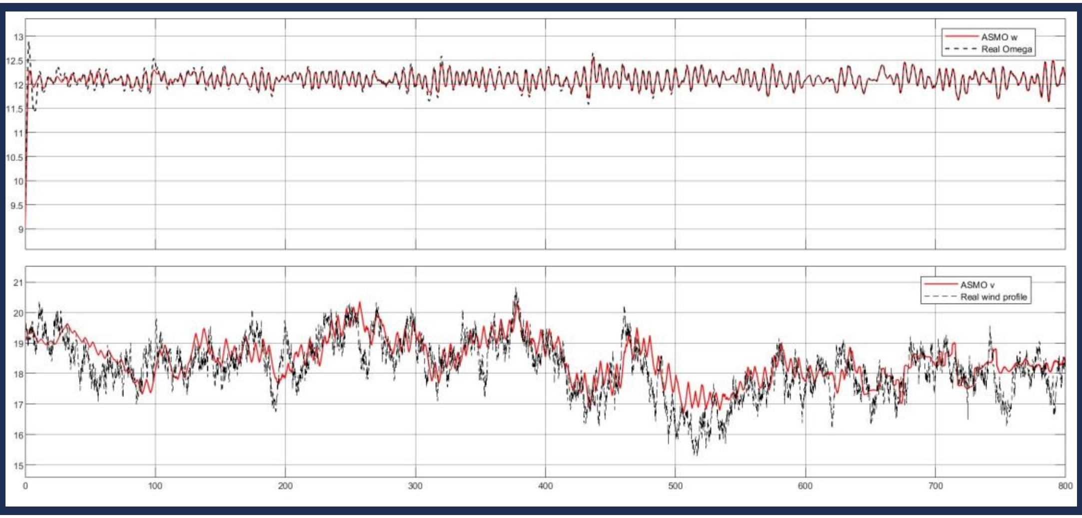
$$\begin{cases} \alpha(t) = k_1(t)|\sigma(t)|^{\frac{1}{2}} + |\psi| + |\Gamma_a| \\ \epsilon(t) = \left[\alpha(t) + |\psi| + w(t - T_e)\right] \cdot T_e + k_2(t) \cdot T_e^2 \\ \dot{\Gamma}_a = k_2(t)\,\text{sgn}\left(\sigma(t)\right), \ \Gamma_a(0) = 0 \end{cases}$$

Results

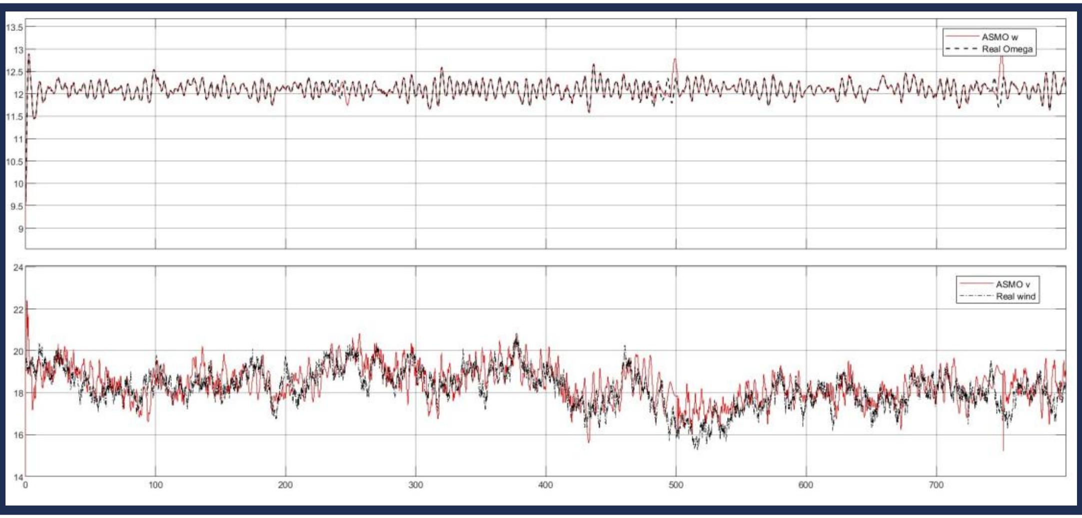
Constant parameter α



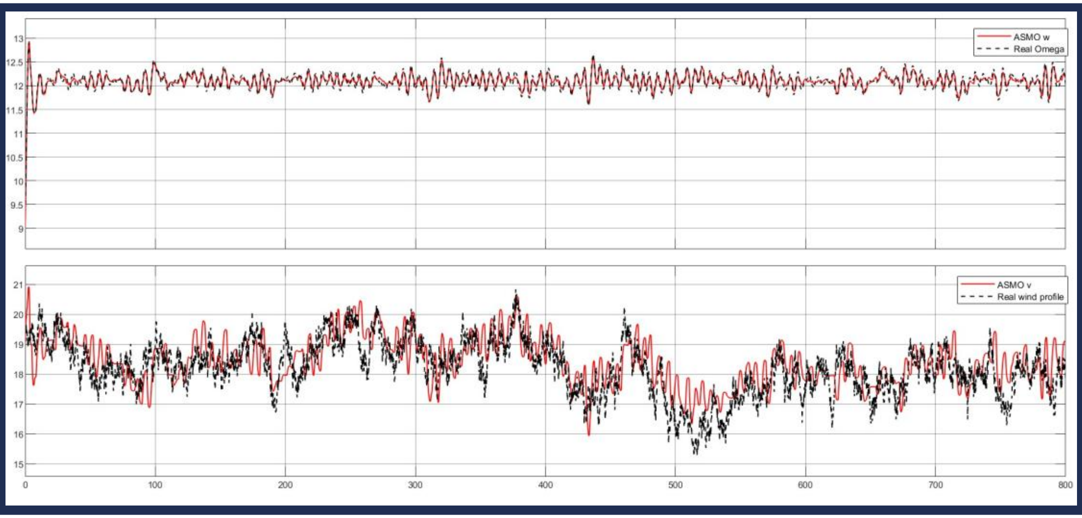
Self-Tuning parameter μ



6 tuning parameters



Auto-Tuning

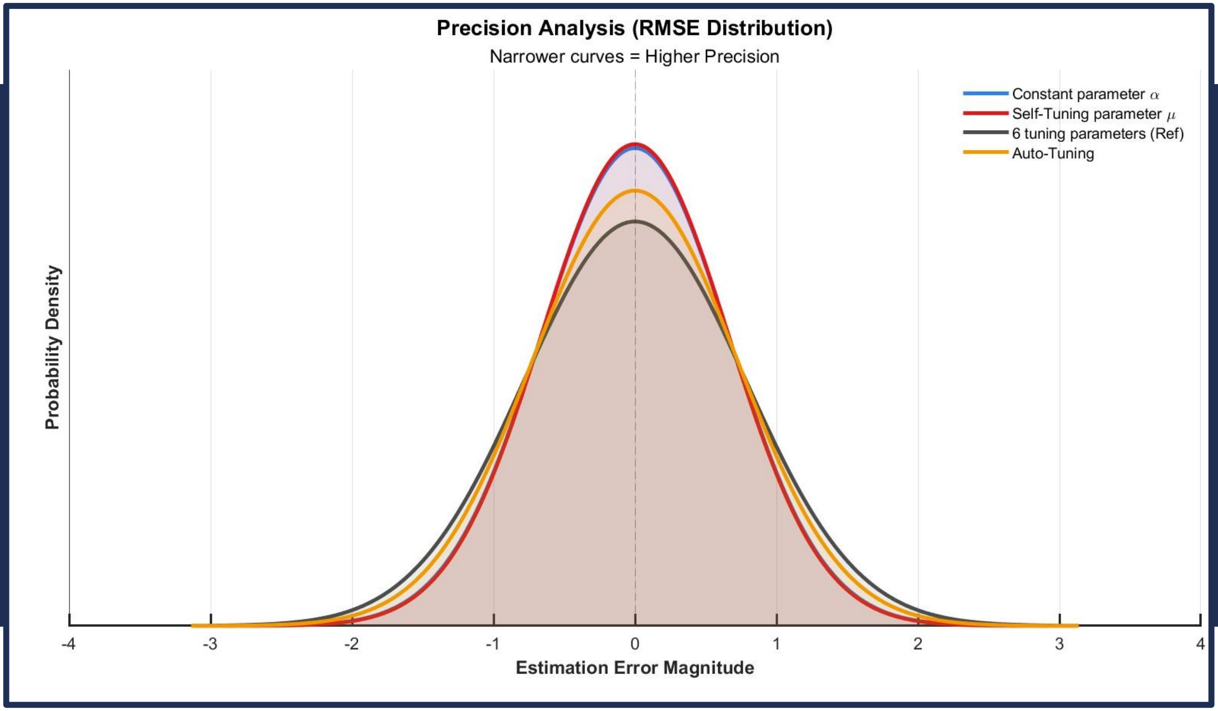


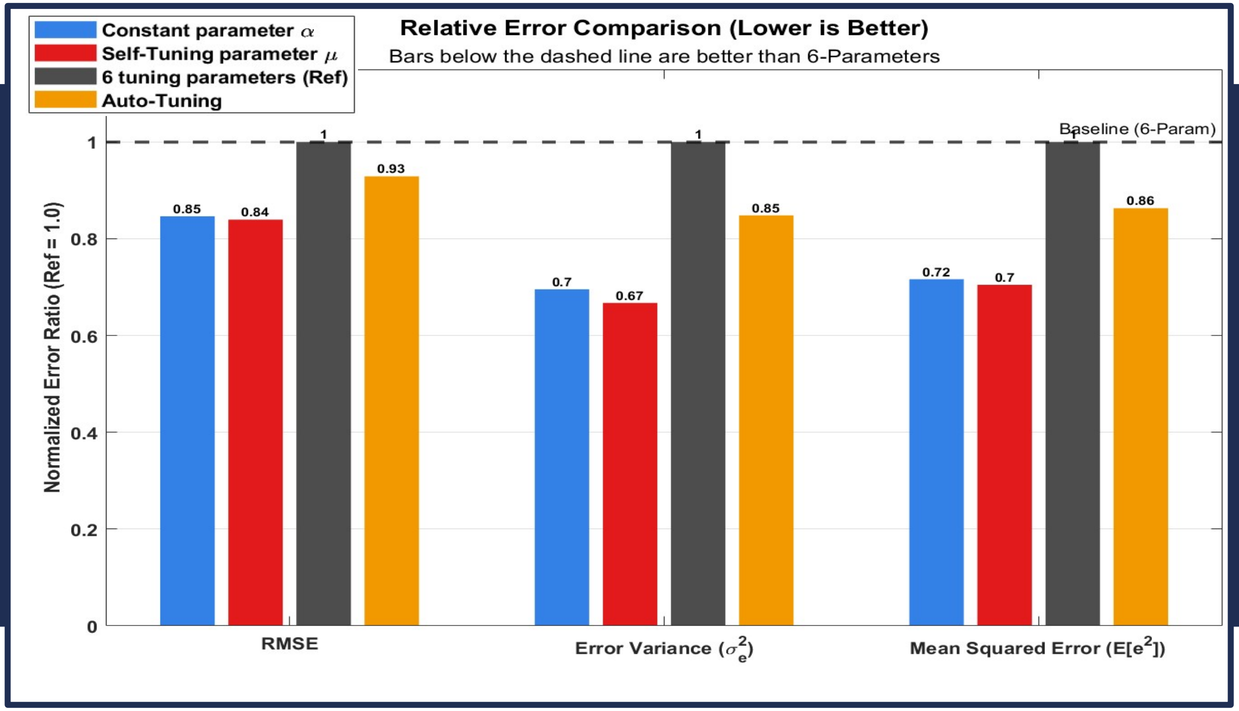
Results

Table I: Time-domain statistical performance metrics for Constant parameter α, Self-Tuning parameter $\bar{\mu}$, 6 tuning parameters, and Auto-Tuning evaluated for rotor speed ω_r and REWS v_r.

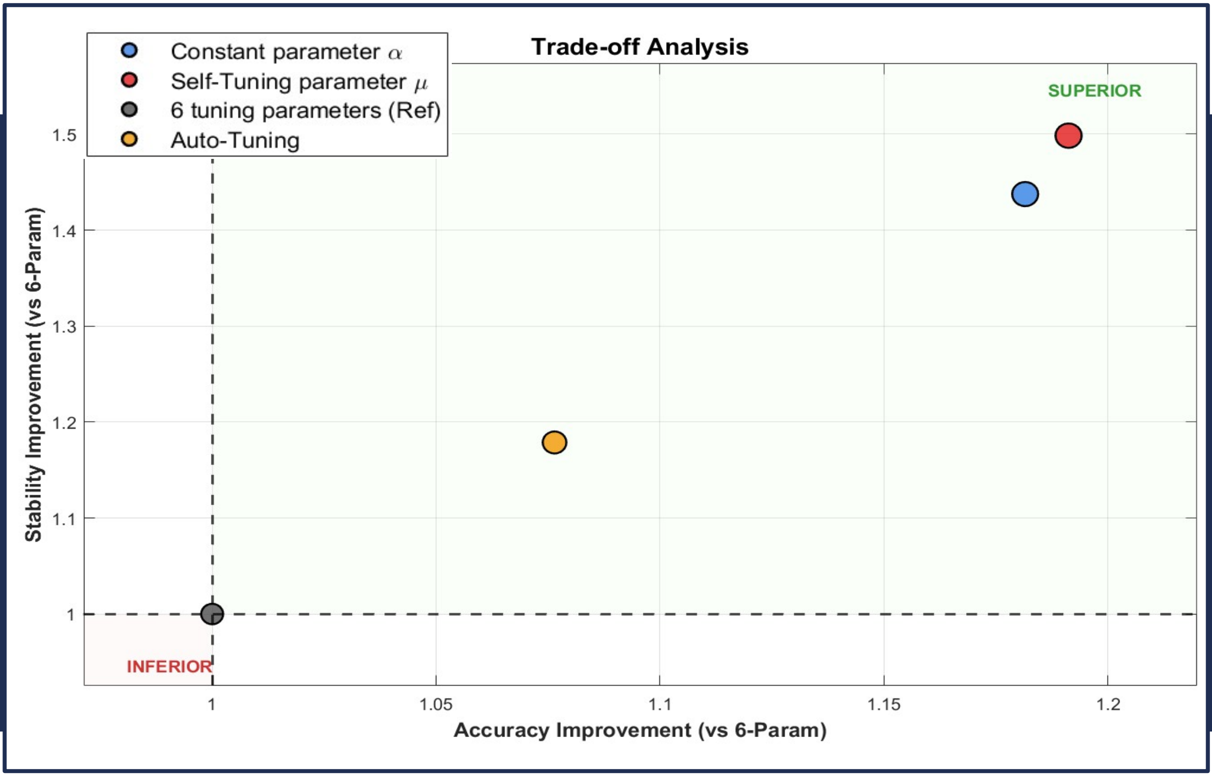
Variable	Algorithm	RMSE	σ_e^2	$E[e^2]$
ω_r	Constant parameter α	$4.022\ x\ 10^{-2}$	$1.612\ x\ 10^{-3}$	$1.618\ x\ 10^{-3}$
	Self-Tuning parameter $\bar{\mu}$	$1.210\ x\ 10^{-2}$	$1.460\ x\ 10^{-4}$	$1.464\ x\ 10^{-4}$
	6 tuning parameters	$8.189\ x\ 10^{-2}$	$6.688\ x\ 10^{-3}$	$6.707\ x\ 10^{-3}$
	Auto-Tuning	$6.619\ x\ 10^{-2}$	$4.380\ x\ 10^{-3}$	$4.381\ x\ 10^{-3}$
v_r	Constant parameter α	$6.638 x\ 10^{-1}$	$3.890\ x\ 10^{-1}$	$4.406\ x\ 10^{-1}$
	Self-Tuning parameter $\bar{\mu}$	$6.584 x\ 10^{-1}$	$3.732 x\ 10^{-1}$	$4.335 x\ 10^{-1}$
	6 tuning parameters	$7.843 x\ 10^{-1}$	$5.592 x\ 10^{-1}$	$6.151 x\ 10^{-1}$
	Auto-Tuning	$7.286 x\ 10^{-1}$	$4.744 x\ 10^{-1}$	$5.309 x\ 10^{-1}$

Results


Precision Analysis (RMSE Distribution)
Narrower curves = Higher Precision
Constant parameter α
Self-Tuning parameter μ
6 tuning parameters (Ref)
Auto-Tuning
Probability Density
-4
-3
-2
-1
0
1
2
3
4
Estimation Error Magnitude


Constant parameter α
Self-Tuning parameter μ
6 tuning parameters (Ref)
Auto-Tuning
Relative Error Comparison (Lower is Better)
Bars below the dashed line are better than 6-Parameters
Baseline (6-Param)
Normalized Error Ratio (Ref = 1.0)
0.85
0.84
1
0.93
0.7
0.67
1
0.85
0.72
0.7
1
0.86
RMSE
Error Variance (σ_e^2)
Mean Squared Error (E[e^2])

Conclusion



Superiority of Self-Tuning: The Self-Tuning Parameter ($\bar{\mu}$) law outperformed all other methods, achieving the lowest RMSE ($6.584x\ 10^{-1}$) and error variance, demonstrating exceptional accuracy and chattering reduction.

Robustness of Auto-Tuning: The Auto-Tuning method proved to be a robust "safe choice," offering consistent improvements over the constant gain baseline without the instability risks of manual tuning.

Risk of Over-Parameterization (Complexity ≠ Accuracy): The 6-Tuning Parameters approach yielded the highest error ($7.843x\ 10^{-1}$), confirming that increased complexity does not always yield better estimation performance and can lead to overfitting.

Future work

Robustness Analysis: Test the observer under more aggressive environmental conditions, including extreme turbulence intensities and severe platform pitching motion.

Parameter Sensitivity: Investigate the observer's resilience to aerodynamic model mismatches (e.g., C_p table uncertainty) and variations in turbine mass/inertia.

Fusion Technique:

A) Investigate fusing of two ASTW algorithms for optimum estimation.

b) Investigate fusing the SOSMO estimator with LiDAR data to create a robust hybrid system that combines the high bandwidth of the observer with the preview capability of LiDAR.

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Thank You